

MISSOURI S&T
Department of Mathematics & Statistics
Stat 3113 - Sections 101 & 102
Exam 1

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Student Name: _____ *Key!*

Good luck and happy reading work!

Instructions:

1. Be sure to print your first and last name in the spaces provided at the top of this page.
 2. Work all problems. **Show all work.** Full credit will be given only if work is shown which *fully justified* your answer.
 3. There should be sufficient space under each problem in which to show your work. **Do not use the back of the page where the problem is on.**
 4. There are **7 problems**. Point values are indicated (in parentheses) for each problem. The total number of points is 100.
 5. Turn off your cell phone.
 6. Calculator is allowed in this Exam.
 7. There are 11 pages in this Exam 1 set (including this page). **Please check to make sure you have 11 pages with 7 problems.**
 8. Please ask only your instructor if you have any questions during the exam.
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Distribution of Marks

Question	Points	Score
1	12	
2	12	
3	12	
4	15	
5	10	
6	19	
7	20	
Total:	100	

1. (12 points) *For the multiple choice questions, shade the correct answer.*

(a) (4 points) The weight (in grams) of cereal dispensed by a machine into cereal boxes is a _____ variable.

☐ Discrete

☐ Ordinal

☐ Nominal

☒ Continuous

(b) (4 points) A and B are two events such that $P(A) = 0.3$, $P(B) = 0.55$, and $P(A \cup B) = 0.65$. Which of the following statements is true:

☐ A and B are mutually exclusive events.

☐ A and B are exhaustive events.

☒ A and B are dependent events.

☐ A and B are independent events.

(c) (4 points) A company recorded daily sales over a week: 15, 8, 35, 12, 30, 20. What is the median number of units sold?

☐ 22.5

☐ 15

☒ 17.5

☐ 18

2. (12 points) Analyze the following scenario and fill in the table below with the corresponding elements: **Individual, Variable, Population, Parameter, Sample, and Statistic.**

You are an engineer conducting a survey to determine the proportion of homeowners in a city who currently use traditional heating systems and would be interested in installing a solar heating system if a rebate were offered. You distribute 2500 questionnaires and receive 1000 responses from homeowners who currently use traditional heating systems.

Label	Description
<i>Individual</i>	A homeowner who currently uses a traditional heating system.
<i>Sample</i>	The 1000 respondents in the city using traditional heating systems.
<i>Population</i>	All homeowners in the city using traditional heating systems.
<i>Statistic</i>	Proportion of respondents interested in installing a solar heating system.
<i>Variable</i>	Whether or not interested in solar heating with a rebate.
<i>Parameter</i>	Proportion of all homeowners with traditional heating interested in solar heating.

3. (12 points) **For the problems below, write the letter from the list that best matches the description. Note: Each answer may be used once, more than once, or not at all.**

- | | | |
|------------------------|--------------------|-----------------------------|
| A Observational study | E Bias group | I Factor |
| B Response variable | F Lurking variable | J Statistically significant |
| C Confounding variable | G Treatment | K Control group |
| D Experimental study | H Random Study | L Replication |

- (a) An *A* involves observing subjects in their natural environment without attempting to alter or control the conditions.
- (b) When a study involves randomly assigning participants to different conditions to assess the impact of interventions, it is an *D*.
- (c) A *F* is a variable that influences both the explanatory and response variables, creating a misleading association between them.

- (d) A G is a specific experimental condition or intervention applied to subjects to observe its effects.
- (e) The B is the outcome measure that researchers aim to understand or predict based on the impact of other variables.
- (f) A K serves as a baseline group in an experiment, receiving no treatment or a standard treatment for comparison with experimental groups.

4. (15 points) **Memory Recall Times Analysis.**

In a memory recall study, subjects were shown a sequence of stimulus words on a computer screen. For each word, they were instructed to recall either a pleasant or unpleasant memory associated with it. Successful recall was recorded when the subject pressed a bar on the keyboard. The dataset consists of the recall times (in seconds) for twenty pleasant and twenty unpleasant memories.

The main focus of this study was to determine if pleasant memories could be recalled more easily and quickly compared to unpleasant ones.

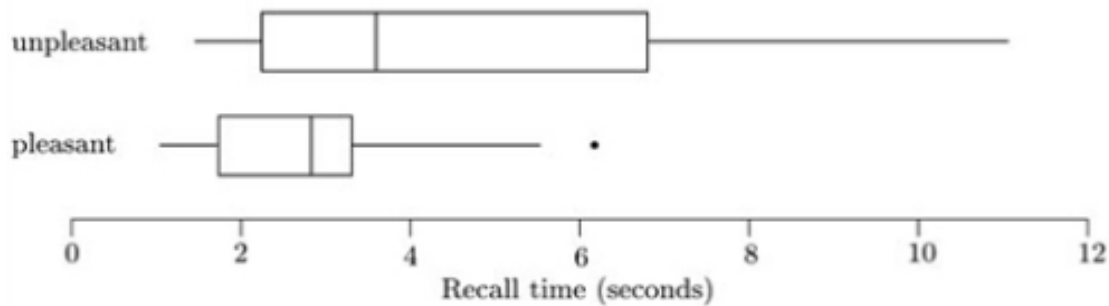


Figure 1: Comparative boxplots of memory recall times.

Refer to the boxplots in Figure 1. Indicate whether each statement is True or False.

- (a) ☐ True ☒ False The median recall time is shorter for unpleasant memories than pleasant memories.
- (b) ☐ True ☒ False Based on the interquartile ranges, the variability in recall times is higher for pleasant memories than for unpleasant ones.
- (c) ☒ True ☐ False The 'unpleasant' sample is right-skewed, with a longer right whisker and a longer segment from the median to the upper quartile.
- (d) ☐ True ☒ False The 'pleasant' sample's skewness pattern, characterized by a longer upper whisker and a shorter upper box segment compared to the lower, indicates that the distribution is symmetric.
- (e) ☐ True ☒ False The boxplot shows one significant outlier in the 'pleasant' sample that is much higher than the upper whisker values and should be removed from the data.

Source: Adapted from "Interpreting data: boxplots and tables", The Open University, 2016.

5. (10 points) A system consists of four components connected as shown in Figure 2.

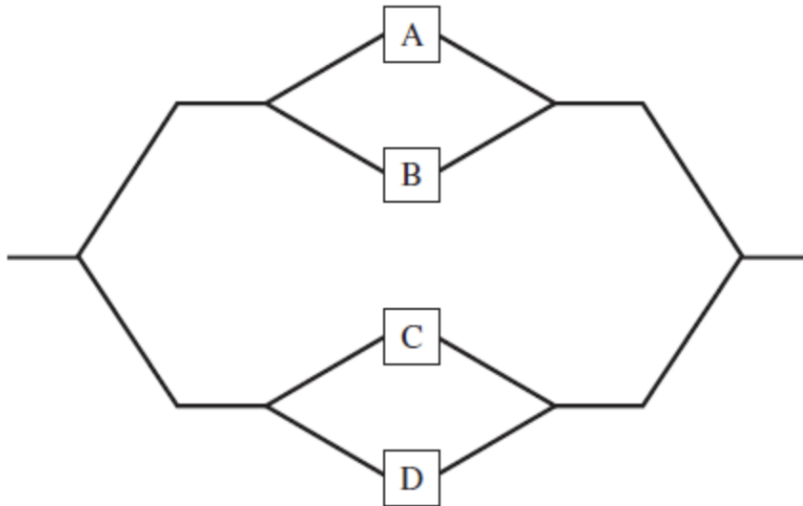


Figure 2: System Configuration.

Assume A , B , C , and D function independently. The probabilities that A , B , C , and D function are 0.88, 0.74, 0.94, and 0.62, respectively.

What is the probability that the entire system works? Round your answer to four decimal places.

- Let A , B , C and D be the events that the components A , B , C and D function respectively.
- Hence, the event that the system functions is $(A \cup B) \cap (C \cup D)$.
- Hence, $IP(\text{system works}) = IP((A \cup B) \cap (C \cup D))$
 - $IP(A \cup B) = 0.88 + 0.74 - 0.88 \times 0.74 = 0.9688$
 - $IP(C \cup D) = 0.94 + 0.62 - 0.94 \times 0.62 = 0.9772$
 - $IP((A \cup B) \cap (C \cup D)) = 0.9688 \times 0.9772 = 0.9468$

Answer = 0.9468

6. (19 points) Of the travelers arriving at a small airport:

- 60% fly on major airlines,
- 30% fly on privately owned planes,
- The remainder fly on commercially owned planes not belonging to a major airline.

Of those traveling on major airlines, 50% are traveling for business reasons. Similarly, 60% of those arriving on private planes and 90% of those arriving on other commercially owned planes are traveling for business reasons.

Let:

$A_1 = \{\text{a person travels on a major airline}\}$ $IP(A_1) = 0.6$

$A_2 = \{\text{a person travels on a privately owned plane}\}$ $IP(A_2) = 0.3$

$A_3 = \{\text{a person travels on a commercially owned plane not belonging to a major airline.}\}$ $IP(A_3) = 0.1$

$B = \{\text{a person travels for business reasons}\}$

Note: one travels either for business or for pleasure.

(a) (7 points) What is the probability that the person is traveling for business?

$$\begin{aligned}
 IP(B) &= IP(B) \\
 &= \sum_{i=1}^3 IP(B|A_i) \cdot IP(A_i) \\
 &\quad \text{Law of Total Probability} \\
 &= (0.5 \times 0.6 + 0.6 \times 0.3 + 0.9 \times 0.1) \\
 &= (0.3 + 0.18 + 0.09)
 \end{aligned}$$

Answer =

0.57

(b) (6 points) What is the probability that the person is traveling for business AND on a commercially owned plane not belonging to a major airline?

$$\begin{aligned}
 IP(B \cap A_3) &= IP(B|A_3) \cdot IP(A_3) \\
 &= 0.9 \times 0.1
 \end{aligned}$$

Answer =

0.09

- (c) (6 points) What is the probability that the person arrived on a commercially owned plane not belonging to a major airline., GIVEN THAT the person is traveling for business reasons?

$$P(A_3|B) = \frac{P(B \cap A_3)}{P(B)}$$
$$= \frac{0.09}{0.57}$$

Answer = 0.16

7. (20 points) Let A and B be defined as follows:

$A = \{\text{American engineering firm will establish a branch office in Toronto}\}$

$B = \{\text{American engineering firm will establish a branch office in Mexico City}\}$

The probability that a branch office is established in Toronto is 0.6, while the probability that a branch office is established in Mexico City is 0.5. The probability that both cities get a branch office is 0.3.

(a) (5 points) What is the probability that Mexico City gets a branch office AND Toronto does not?

$$\begin{aligned} IP(A^c \cap B) &= IP(B) - IP(A \cap B) \\ &= 0.5 - 0.3 \end{aligned}$$

Answer =

0.2

(b) (5 points) What is the probability that a branch office will be established in AT LEAST one of the two cities?

$$\begin{aligned} IP(A \cup B) &= IP(A) + IP(B) - IP(A \cap B) \\ &= 0.6 + 0.5 - 0.3 \end{aligned}$$

Answer =

0.8

(c) (5 points) GIVEN THAT Toronto does not get a branch office, what is the probability that Mexico City does get a branch office?

$$\begin{aligned} IP(B^c | A^c) &= \frac{IP(A^c \cap B^c)}{IP(A^c)} = \frac{IP((A \cup B)^c)}{IP(A^c)} \\ &= \frac{1 - IP(A \cup B)}{1 - IP(A)} \\ &= \frac{1 - 0.8}{1 - 0.6} \end{aligned}$$

Answer =

0.5

(d) (5 points) Are events A and B independent? Justify your answer.

$$\bullet P(A \cap B) = 0.3$$

$$\bullet P(A) \times P(B) = 0.6 \times 0.5 = 0.3$$

Answer =

yes